

Constant-Wind Zermelo Navigation Beyond the Randers Chamber

Route-A source-local certificate HCS-C305

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Abstract

For every finite $d \geq 1$, constant wind $W \in \mathbb{R}^d$, control cap $c \geq 0$, and target, we give the complete minimum-time atlas for $\dot{x} = W + u$, $|u| \leq c$. A single theorem covers weak, critical, and strong wind, including exact reachability, value formulas, and all attainable times. It also proves the unique constant optimizer, HJB equation, symmetries, Mach-cone regularity, and all zero-data faces.

1 The all-chamber value theorem

Consider measurable controls u satisfying $|u(t)| \leq c$ a.e. and

$$\dot{x} = W + u, \quad x(0) = 0. \quad (1)$$

For a target y , write

$$w = |W|, \quad p = W \cdot y, \quad r = |y|, \quad a = w^2 - c^2, \quad D = p^2 - ar^2.$$

Theorem 1 (Complete constant-wind atlas). *At exact time $t \geq 0$, the reachable set is*

$$\mathcal{R}(t) = Wt + ct\bar{B}. \quad (2)$$

For $y \neq 0$, minimum time and all attainable times are as follows.

1. *If $w < c$, every y is reachable,*

$$T(y) = \frac{\sqrt{p^2 + (c^2 - w^2)r^2} - p}{c^2 - w^2}, \quad \mathcal{I}(y) = [T, \infty). \quad (3)$$

2. *If $w = c > 0$, reachability is equivalent to $p > 0$, and then*

$$T(y) = \frac{r^2}{2p}, \quad \mathcal{I}(y) = [T, \infty). \quad (4)$$

3. *If $w > c$, reachability is equivalent to $p > 0$ and $D \geq 0$. Then*

$$T_{\mp}(y) = \frac{p \mp \sqrt{D}}{w^2 - c^2}, \quad T(y) = T_-(y), \quad \mathcal{I}(y) = [T_-, T_+]. \quad (5)$$

For every nonzero reachable target the time-optimal control is unique up to null sets and constant:

$$u_*(s) = y/T(y) - W, \quad |u_*| = c. \quad (6)$$

Proof. Integrating (1) shows that control averages fill $c\bar{B}$, proving (2). Thus y is attainable at t exactly when

$$|y - Wt|^2 \leq c^2 t^2 \iff f(t) = at^2 - 2pt + r^2 \leq 0. \quad (7)$$

For $y \neq 0$, $f(0) > 0$. If $a < 0$, its roots have opposite signs and the positive root is (3); the inequality holds thereafter. If $a = 0$, it is linear and crosses zero at positive time exactly when $p > 0$, giving (4). If $a > 0$, it has a nonnegative-time sublevel interval exactly when $p > 0$ and $D \geq 0$. Both roots are then positive and the sublevel set is precisely (5); in particular the minimum is the smaller root, not T_+ .

At first contact, $|y - WT| = cT$. Any optimal control obeys

$$\left| T^{-1} \int_0^T u(s) ds \right| = c \leq T^{-1} \int_0^T |u(s)| ds \leq c.$$

Equality in the strictly convex Euclidean triangle inequality forces the control to equal $y/T - W$ a.e. For $c = 0$ this conclusion is immediate. $\square$

2 Value geometry, HJB, and exact boundaries

Proposition 1 (Symmetries and HJB). *For every orthogonal Q , $\lambda > 0$, and $s > 0$,*

$$T_{QW,c}(Qy) = T_{W,c}(y), \quad T_{W,c}(\lambda y) = \lambda T_{W,c}(y), \quad T_{sW,sc}(y) = s^{-1}T_{W,c}(y). \quad (8)$$

On each interior component where T is finite and $y \neq 0$, it is smooth and

$$W \cdot \nabla T + c|\nabla T| = 1. \quad (9)$$

Proof. Equation (8) follows from (1), or directly from p, r, w, c . First assume $c > 0$. At first contact put $F(y, T) = |y - WT|^2 - c^2 T^2 = 0$. Implicit differentiation gives

$$\nabla T = \frac{y - WT}{p - (w^2 - c^2)T}. \quad (10)$$

The denominator is $\sqrt{D} > 0$ on the selected smooth radical branch and is $p > 0$ in critical wind. Using $|y - WT| = cT$ in (10) yields (9) with the displayed plus sign. If $c = 0$ and $d > 1$, the reachable ray has empty Euclidean interior. If $c = 0$, $d = 1$, and $W \neq 0$, write $e = W/|W|$ and $y = se$ on the open forward ray. Then $T = s/|W|$, $\nabla T = e/|W|$, and $W \cdot \nabla T = 1$ directly. $\square$

The formulas remain exact on their stated faces, but the domains change.

Face	Exact conclusion
$y = 0$	$T = 0$; every $t \geq 0$ is feasible if $w \leq c$, only $t = 0$ if $w > c$.
$W = 0 < c$	$T(y) = y /c$ for every target.
$c = 0, W \neq 0$	only $y = tW$, $t \geq 0$, is reachable; $u = 0$ is forced.
$W = c = 0$	only the zero target is reachable.
$w = c > 0$	nonzero finite-value domain is the open half-space $p > 0$.
$0 < c < w$	full finite-value domain is the closed cone $\{0\} \cup \{y \neq 0 : p > 0, D \geq 0\}$.
$D = 0$	double time root and square-root loss on a nontrivial cone boundary.

In one dimension with $0 < c < w$, the nonzero reachable ray lies in the smooth interior algebraically; a nontrivial angular cone boundary requires higher dimension. When $c = 0$, the ray is handled separately. Weak wind is the gauge of a velocity ball containing the origin, but the strong-wind value is not a global Finsler norm because it is infinite outside its cone.

Source lineage

Zermelo's 1931 paper is the historical owner token for the navigation problem. Bao, Robles, and Shen provide navigation/Randers geometric context. The citations establish lineage, not novelty or priority.

References

- [1] E. Zermelo, "Über das Navigationsproblem bei ruhender oder veränderlicher Windverteilung," *ZAMM* 11 (1931), 114–124. DOI: [10.1002/zamm.19310110205](https://doi.org/10.1002/zamm.19310110205).
- [2] D. Bao, C. Robles, and Z. Shen, "Zermelo navigation on Riemannian manifolds," *J. Differential Geom.* 66 (2004), 377–435. DOI: [10.4310/jdg/1098137838](https://doi.org/10.4310/jdg/1098137838).